

Superiority of Lateral Decubitus Strategy in Preventing Atelectasis From Obscuring Targets During Robotic Bronchoscopy

Lateral Decubitus Strategy vs Ventilatory Strategy to Prevent Atelectasis Trial



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BACKGROUND: Atelectasis during peripheral bronchoscopy can cause CT-to-body divergence, can cause false-positive radial-probe endobronchial ultrasound images, and can obscure a target. As shown in prior studies, ventilatory strategies to prevent atelectasis, although useful, cannot completely avoid this phenomenon.

RESEARCH QUESTION: Is a lateral decubitus superior to a ventilatory strategy in preventing atelectasis from obscuring targets in patients with lung nodules in dependent zones?

STUDY DESIGN AND METHODS: This was a randomized controlled study (1:1) in which patients undergoing robotic bronchoscopy for nodules ≤ 3 cm in dependent lung zones were randomized to lateral decubitus strategy (LADS) vs a ventilatory strategy to prevent atelectasis (VESPA). Patients who developed atelectasis obscuring the target crossed over to the opposite arm. Primary outcome was the development of atelectasis obscuring the target detected with mobile cone-beam CT scan. Secondary end points included time to lesion (TIL), diagnostic yield, and safety.

RESULTS: A total of 62 patients were analyzed: 29 in LADS and 33 in VESPA. No patients developed atelectasis obscuring target in the LADS group, and 9 patients did in the VESPA group (27.3%; 95% CI, 13.3%-45.5%; $P = .002$). TIL was achieved in 29 patients (100%; 95% CI, 88.1%-100%) in the LADS group and in 24 patients (72.7%; 95% CI, 54.5%-86.7%) in the VESPA group ($P = .005$). Diagnostic yield on index biopsy was made in 25 patients (86.2%; 95% CI, 68.3%-96.1%) in the LADS group and 19 patients (57.6%; 95% CI, 39.2%-74.5%) in the VESPA group ($P = .026$). In all 9 patients who crossed over from the VESPA group to the LADS group, atelectasis was eliminated leading to TIL, and diagnosis was obtained in 7 (78%). There were no major complications and no differences in the rate of minor complications.

INTERPRETATION: Our results suggest that LADS is superior to VESPA in preventing atelectasis from obscuring targets in patients with nodules located in dependent areas, can help eliminate atelectasis, is safe, and is also associated with better procedural outcomes such as TIL and diagnostic yield.

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FOR EDITORIAL COMMENT, SEE PAGE 880

Take-Home Points

Research Question: Is a lateral decubitus superior to a ventilatory strategy in preventing atelectasis from obscuring targets in patients with lung nodules in dependent zones?

Results: We randomized patients undergoing robotic bronchoscopy for small nodules in dependent lung zones to lateral decubitus strategy (LADS) vs a ventilatory strategy to prevent atelectasis (VESPA) and found no patients with atelectasis obscuring target in the LADS group vs 9 patients in the VESPA group (27.3%; 95% CI, 13.3%-45.5%; $P = .002$), tool in lesion in all 29 patients in the LADS group vs 24 patients (72.7%) in the VESPA group ($P = .005$), and diagnosis on index biopsy in 25 patients (86.2%) in the LADS group vs 19 patients (57.6%) in the VESPA group ($P = .026$).

Interpretation: Our findings show that LADS is superior to VESPA in preventing atelectasis from obscuring targets in patients with nodules located in dependent areas, can help eliminate atelectasis, and is also associated with better procedural outcomes such as tool in lesion and diagnostic yield.

Atelectasis is a common occurrence during bronchoscopy under general anesthesia (GA), affecting predominantly the dependent lung areas of the lower lobes, and is more common with higher BMI and longer time under GA.¹⁻⁴ It can negatively impact the outcomes of peripheral bronchoscopy by causing false-positive radial-probe endobronchial ultrasound images or completely obscuring targets preventing identification with cone-beam CT (CBCT) scan, eventually leading to nondiagnostic bronchoscopy.¹⁻⁸ In this new era of robotic bronchoscopy, navigation to a target (whether electromagnetic or shape sensing-based) relies on the creation of a virtual bronchoscopy

tree and target identification using planning software, a process based on a preprocedural chest CT scan. The greater the atelectasis, the greater the difference in shape and volume of the lungs between the preprocedural chest CT scan (performed at deep inspiration) and the actual lungs during bronchoscopy, and the greater the distance between the virtual and the actual target during bronchoscopy. This phenomenon is described as CT-to-body divergence, and atelectasis is partly responsible for it.⁶ Although the rapid adoption of 3-dimensional imaging technologies such as CBCT scan or tomosynthesis can help correct for CT-to-body divergence, if the target is significantly obscured, the procedure may be aborted.

Understanding the relevance of this phenomenon, our group previously carried out a multicenter randomized controlled study using a ventilatory strategy to prevent atelectasis (VESPA), which involved a recruitment maneuver, F_{IO_2} titration ($< 100\%$), positive end-expiratory pressure (PEEP) of 8 to 10 cm H_2O , and tidal volume (V_T) of 6 to 8 mL/kg of ideal body weight (IBW).⁹ This strategy was well tolerated, and it substantially decreased the rate of atelectasis from 84% in the control group to 29%. Retrospective reports with other ventilatory strategies (with V_T and PEEP twice as high as those in VESPA, and with prolonged breath-holds) also reduced the atelectasis rate to the 20% to 30% range but caused hemodynamic instability in up to 70% of the patients.^{10,11}

Realizing that these ventilatory strategies were not fully able to prevent atelectasis and, when aggressive, risked hemodynamic instability, our group developed a positional strategy to prevent atelectasis which would require no ventilatory adjustments, the lateral decubitus strategy (LADS). In a brief research letter, we reported our initial experience of 11 patients undergoing LADS during robotic bronchoscopy, showing no atelectasis,

ABBREVIATIONS: ATS = American Thoracic Society; CBCT = cone-beam CT; CHEST = American College of Chest Physicians; DY = diagnostic yield; GA = general anesthesia; IBW = ideal body weight; IQR = interquartile range; LADS = lateral decubitus strategy; m-CBCT = mobile cone-beam CT; PEEP = positive end-expiratory pressure; PP = prone positioning; TIL = tool in lesion; VESPA = ventilatory strategy to prevent atelectasis; V_T = tidal volume

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which led to the current study.¹² We conducted a randomized controlled study to determine if LADS is superior to VESPA in reducing the intraprocedural

development of atelectasis obscuring targets during robotic bronchoscopy under GA in patients with lung nodules located in dependent areas.

Study Design and Methods

Study Setting and Patients

The study was performed at the University of Texas MD Anderson Cancer Center with institutional review board approval (No. 2022-0756). Enrollment started in June 2023 and ended in June 2025. We enrolled adult patients referred to pulmonary medicine for diagnosis of a peripheral lung nodule (outer two-thirds of the lungs) suspicious for malignancy, up to 3 cm in long axis, located in dependent areas of the lung (Fig 1).¹³ For a nodule to be considered in a dependent area, > 50% of its volume had to lie below a horizontal line drawn at the anterior edge of the corresponding vertebral body on axial chest CT scan, matching our previously described zones 1 and 2, areas at high atelectasis

risk (Fig 2).⁴ All patients signed informed consent. We excluded patients with pure ground-glass opacity nodules, patients with evidence on chest CT scan of lung consolidation, pregnant patients, and patients with pleural effusions, ascites, diaphragmatic paralysis, or known severe air-trapping with residual volume > 150%. We also excluded patients with known shoulder injuries on the shoulder that would stay down on lateral decubitus.

Study Design and Procedures

This was a prospective, randomized (1:1), controlled trial comparing the intraprocedural development of atelectasis obscuring the target during robotic bronchoscopy under GA of patients receiving VESPA

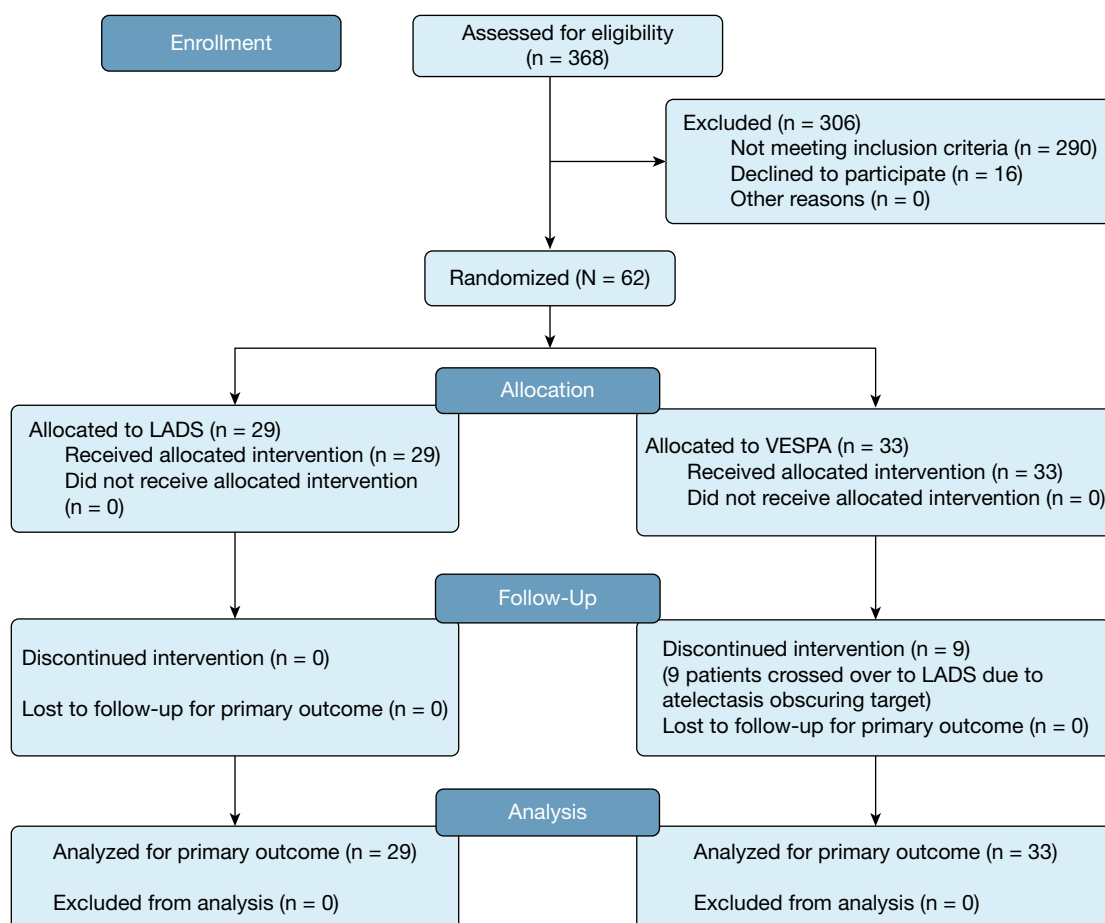


Figure 1 – Consolidated Standards of Reporting Trials flow diagram. LADS = lateral decubitus strategy; VESPA = ventilatory strategy to prevent atelectasis. (Hopewell et al.¹³)



Figure 2 – Definition of dependent lung zones at risk of atelectasis.

(control group) vs LADS (treatment group) strategy. Randomization was performed immediately before bronchoscopy; it was computer-generated and stratified according to BMI (> 30 vs ≤ 30 kg/m²) and tumor location (upper lobe vs lower lobe). If atelectasis were found to be obscuring the target, the patients crossed over to the opposite arm.

Bronchoscopy was performed following our standard of care with total IV anesthesia including neuromuscular blocking agents in all cases. All patients were intubated in supine position and were ventilated with volume control ventilation or volume control pressure-limited ventilation. A bronchoscopic airway examination and clearance of secretions was performed immediately after intubation and before the recruitment maneuver (VESPA group) or before positioning the patient in lateral decubitus (LADS group). The VESPA group received ventilation through an endotracheal tube, V_T of 6 to 8 mL/kg of IBW, $FiO_2 < 100\%$ (titrated to maintain an oxygen saturation of $> 94\%$), and PEEP of 8 to 10 cm H₂O. In addition, patients in the VESPA group underwent a recruitment maneuver immediately after intubation with 10 consecutive breaths at a plateau pressure of 40 cm H₂O with PEEP of 20 cm H₂O in pressure control mode. Patients in the LADS group were intubated and ventilated with V_T of 6 to 8 mL/kg of IBW, FiO_2 of 100%, and PEEP of 0 to 5 cm H₂O. We assigned a low PEEP to LADS because higher PEEP can cause hemodynamic instability in certain at-risk patients, and we aimed to develop a

strategy that could prevent atelectasis without the need for high PEEP. After intubation and airway examination, patients were turned to the lateral decubitus position with the target lesion side up, similar to a pleuroscopy procedure as previously described.¹² A bean bag and foam straps secured the patient, the dependent arm was flexed 90°, and the nondependent arm was padded alongside (Fig 3). In both groups, the robot (Ion; Intuitive Surgical) was positioned at the head of the bed and the mobile cone-beam CT (m-CBCT) scanner (Cios Spin; Siemens) was positioned on the left of the patient's bed (due to our room configuration). Iso-centering of the m-CBCT C-arm, collision check, and docking of the robot were performed per standard of care. For patients in the LADS group, the view of the carina in the video camera was seen roughly perpendicular to the virtual view of the virtual carina, which was adjusted (90° rotation) during the step of confirmation of main carina. Registration, navigation, and sampling were performed in the same manner as when the patient is traditionally supine. In both groups, when the virtual target was reached, a m-CBCT spin was performed either to use integration (localizing the target in the intraoperative m-CBCT spin, transferring that information back to the robotic platform, and updating the position of the target) or to confirm tool in lesion (TIL) if the needle had been deployed. When m-CBCT scan showed atelectasis obscuring the target, the patient was crossed over to the opposite arm to eliminate atelectasis, localize the target, and continue with the procedure. All patients required a m-CBCT spin to demonstrate TIL before sampling.

Once the target was reached and TIL was confirmed, biopsies of the lesion were performed with transbronchial needle aspiration (≥ 4 samples, per protocol) and a combination of forceps biopsies, cryobiopsies, or cytology brushes, at the discretion of the operator. Rapid on-site examination was used in all cases. When the target lesion was suspected to be a lung malignancy, nodal staging via endobronchial ultrasound-guided (convex-probe endobronchial ultrasound) was performed per our standard of care.

Definition of Atelectasis Obscuring Target and Primary and Secondary End Points

Atelectasis was deemed to be obscuring the target (enough to preclude sampling) when $> 50\%$ of the volume of the target was obscured by the consolidated (atelectatic) parenchyma, assessed with m-CBCT scan read by the bronchoscopist (Fig 4). Atelectasis not obscuring the target was recorded but did not trigger cross over because the target remained visible and sampling was possible.

Figure 3 – Lateral decubitus positioning.



The primary objective was to determine if LADS could reduce the intraprocedural development of atelectasis obscuring the target during robotic bronchoscopy under GA when compared with VESPA. This was determined by comparing the proportion of patients with atelectasis obscuring the target found in the m-CBCT scan in each group. TIL was a secondary end point, and it was defined as any portion of the needle visualized within the target in all 3 axes, assessed by the proceduralist during the procedure via m-CBCT scan. We defined diagnostic yield (DY), also a secondary end point, as the proportion of patients with a malignant diagnosis or specific benign diagnosis—as listed in American Thoracic Society (ATS)/American College

of Chest Physicians (CHEST) consensus—on the index biopsy only, not based on follow-up.¹⁴ If patients developed atelectasis obscuring target, the case was considered as no TIL (needle would not be deployed, or if had been deployed, TIL could not be visualized) and nondiagnostic (samples would not be obtained without showing TIL first, per our standard of care). Analyses for both TIL and DY outcomes were conducted based on the arm to which patients were originally randomized, regardless of the results obtained after any subsequent cross over. Analysis of DY was conducted in a nonblinded fashion, but it adhered strictly to ATS/CHEST guidelines, and all diagnoses were reported in a dedicated table. Intrabronchoscopy and

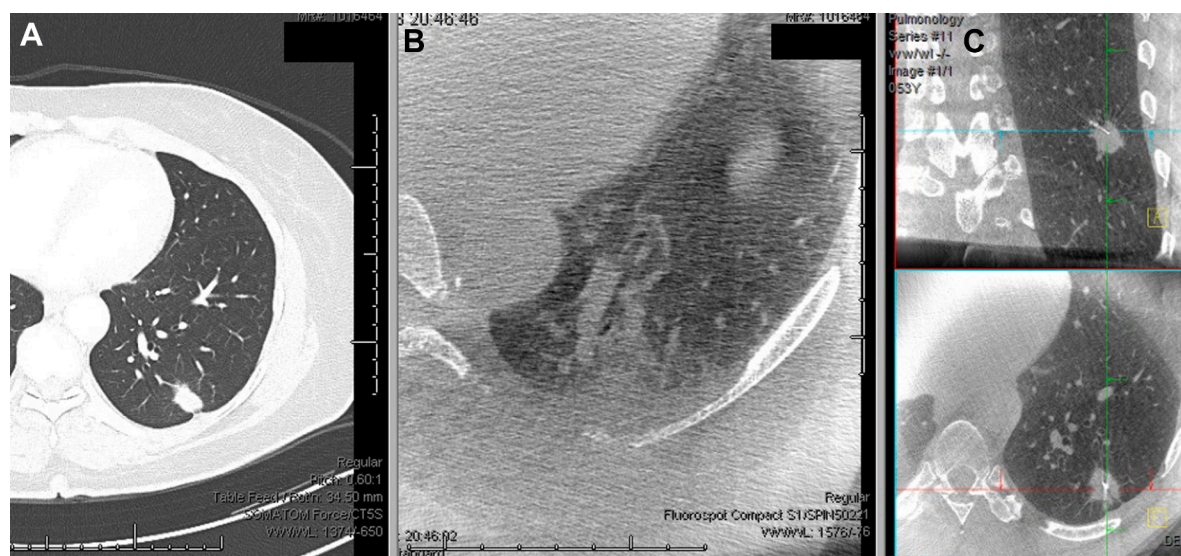


Figure 4 – A-C, Atelectasis obscuring target during VESPA and eliminated with LADS. A, Preprocedural standard chest CT scan showing left lower lobe target. B, Cone-beam CT scan showing atelectasis obscuring > 50% of the target during VESPA. C, Elimination of atelectasis and demonstration of tool in lesion after crossing over to LADS. LADS = lateral decubitus strategy; VESPA = ventilatory strategy to prevent atelectasis.

postbronchoscopy complications after a priori definitions were described and compared, with bleeding recorded per the Nashville Scale.¹⁵

Additional data included demographics, target location/characteristics, procedural details (robotic bronchoscopy time, nodal staging, and sampling tools), and ventilatory settings. Radiation exposure was reported per guidelines.¹⁶ Study data were collected and managed using REDCap hosted at our institution.¹⁷

Statistics

Sample size calculation was based on our prior studies, where the incidence rate of atelectasis during bronchoscopy under GA with VESPA was approximately 29%.⁹ We considered LADS clinically effective if it can decrease the proportion of patients with targets obscured by atelectasis to 1%, consistent with a small cohort of patients we previously published.¹² We required a total of 60 patients (30 patients in each arm) to detect the aforementioned difference and achieve at least 80% power with 5% significance level using 2-sided Fisher exact test. PASS 2005 (NCSS, LLC) was used for the sample size calculation. Of note, after publication of the aforementioned small cohort, we continued to perform robotic bronchoscopy in lateral decubitus position for lesions in dependent areas as part of routine practice, and we did so in a larger number of patients before the current trial. None of those patients developed atelectasis, and this

clinical experience also informed the effect size used in the power calculation.

Descriptive statistics (frequencies, proportions, means, SDs, medians, and interquartile ranges [IQRs]) were used to summarize patient characteristics. The proportion of patients developing atelectasis obscuring target by m-CBCT scan and its 95% CI were estimated in each arm. Proportions were compared using Fisher exact test to accomplish primary objective. Logistic regression models were used to evaluate the effect of covariate on developing atelectasis obscuring target. We included the following covariates which we thought would be clinically relevant: age, BMI, target size, sex, smoking history, COPD, history of malignancy, target location (upper lobe/lower lobe), CT appearance, and atelectasis zone. Categorical variables were summarized using descriptive statistics and compared between arms using the χ^2 test or Fisher exact test. Continuous variables were summarized using descriptive statistics and compared between arms using 2-sample *t* test or Wilcoxon rank-sum test. Bonferroni correction was applied to control for multiple comparisons among 2 secondary end points. Observed complications were tabulated by study arm. *P* < .05 was considered statistically significant. All analyses were performed using SAS 9.4 (SAS Institute).

Study Registration

This study is registered at [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT05714033) as NCT05714033.

Results

A total of 62 patients were included in the analysis: 29 in LADS and 33 in VESPA (Fig 1). There were no statistically significant differences in patients and targets characteristics between the 2 arms (Table 1). In the entire cohort, the median BMI was 27.9 kg/m² (IQR, 23.8-32.7), the median target size was 1.6 cm (IQR, 1.1-2.1), the median distance to the pleura was 0.6 cm (IQR, 0-1.4), 87.1% of the targets were in the lower lobes, and only 22.6% had a bronchus sign. Procedural characteristics are depicted in Table 2. Besides the expected differences in ventilatory parameters, time from intubation to robot docking was longer in the LADS group with a median of 18 (IQR, 15-28) vs 10 minutes (IQR, 8-15) in the VESPA group (*P* < .001). Overall, the median fluoroscopy time was 8.1 minutes (IQR, 6.5-9.9), the median number of spins was 3 (IQR, 2-5), and the median dose-area product was 57.6 Gy·cm² (IQR, 43.5-101.4), with no differences between groups. Transbronchial needle aspiration was used in 100% of the cases, cryobiopsies was used in 45.2%,

cytology brush was used in 8.1%, and forceps biopsy was used in 6.5%.

For the primary objective, no patients developed atelectasis obscuring the target in the LADS group and 9 patients (27.3%; 95% CI, 13.3%-45.5%) did in the VESPA group (*P* = .002). These 9 patients crossed over to the LADS group. Analysis demonstrated TIL in all 29 patients (100%; 95% CI, 88.1%-100%) in the LADS group and in 24 of 33 patients (72.7%; 95% CI, 54.5%-86.7%) in the VESPA group (*P* = .005). DY on index biopsy was made in 25 of 29 patients (86.2%; 95% CI, 68.3%-96.1%) in the LADS group and in 19 of 33 patients (57.6%; 95% CI, 39.2%-74.5%) in the VESPA group (*P* = .026). Diagnosis in each group is available in Table 3. Atelectasis was resolved completely in all patients who crossed over from the VESPA group to the LADS group, leading to TIL in all 9 patients (100%), and diagnosis in 7 of 9 patients (78%). Atelectasis not obscuring the target was found in 3 of 33 patients

TABLE 1] Patient and Target Characteristics

Patient and Target Characteristics	VESPA Group (n = 33)	LADS Group (n = 29)
Age, y	68.0 (61.0-75.0)	69.0 (64.0-72.0)
BMI, kg/m ²	25.6 (23.1-32.8)	28.6 (25.8-32.0)
Sex		
Female	20 (60.6)	16 (55.2)
Male	13 (39.4)	13 (44.8)
Smoking		
Current	5 (15.2)	5 (17.2)
Former	11 (33.3)	16 (55.2)
Never	17 (51.5)	8 (27.6)
COPD	1 (3.0)	5 (17.2)
ECOG		
0	15 (45.5)	11 (37.9)
1	16 (48.5)	16 (55.2)
2	2 (6.1)	1 (3.4)
3	0 (0.0)	1 (3.4)
ASA		
1	1 (3.0)	0 (0)
2	7 (21.2)	3 (10.3)
3	25 (75.8)	26 (89.7)
History of previous malignancy	17 (51.5)	18 (62.1)
Target size, cm	1.7 (1.2-2.2)	1.3 (1.1-1.8)
Target location		
Upper lobes	4 (12.1)	4 (13.8)
Lower lobes	29 (87.9)	25 (86.2)
Distance to pleura, cm	0.7 (0.0-1.3)	0.5 (0.0-1.4)
Atelectasis zone		
Zone 1	11 (33.3)	12 (41.4)
Zone 2	22 (66.6)	17 (58.6)
Bronchus sign	8 (24.2)	6 (20.7)
CT scan appearance		
Solid	28 (84.8)	23 (79.3)
Semisolid	5 (15.2)	6 (20.7)

Data are presented as No. (%) or median (interquartile range). ASA = American Society of Anesthesiology; ECOG = Eastern Cooperative Oncology Group; LADS = lateral decubitus strategy; VESPA = ventilatory strategy to prevent atelectasis.

(9.1%) in the VESPA group and none in the LADS group. Univariate logistic regression models assessing predictors of atelectasis obscuring target within the VESPA arm only found higher BMI associated with increased odds of experiencing atelectasis obscuring target (1-unit change in BMI had an OR of 1.21; 95% CI, 1.03-1.42; $P = .018$).

There were no deaths, no major complications, and no differences in the rate of minor complications. Two patients had grade II bleeding in the LADS group, 1 patient had pneumothorax in each arm, and 3 patients in each arm required a single dose of vasopressors.

Discussion

We report the first randomized controlled trial to our knowledge comparing a new positional strategy (LADS) with an established ventilatory strategy (VESPA) for preventing atelectasis from obscuring targets localized in dependent areas of the lungs during robotic bronchoscopy under GA. Our study demonstrated that LADS was superior to VESPA in preventing atelectasis from obscuring targets, translating into superior procedural outcomes such as TIL and DY. Furthermore, LADS fully eliminated atelectasis in patients initially undergoing VESPA after cross over.

Unlike prior studies on atelectasis risk reduction during bronchoscopy, we used a more clinically relevant end point, atelectasis obscuring the target, rather than overall atelectasis rate. If atelectasis occurs but it is not obscuring the target, even when causing CT-to-body divergence, 3-dimensional imaging such as CBCT scan can easily correct for it and still obtain TIL. However, if atelectasis prevents us from visualizing the target with CBCT scan, then TIL cannot be confirmed, and sampling should not be attempted, or it would be done blindly. Following this rationale, also unlike prior studies, we only included patients with nodules in dependent areas, in essence, those at high risk of being obscured by atelectasis. LADS outperformed VESPA with 0 patients developing atelectasis obscuring targets and fully eliminating atelectasis in the 27% of patients in the VESPA group whose targets were obscured, leading to TIL in all the cases who crossed over to the LADS group. In the VESPA group, the inability to visualize targets translated into poorer TIL and DY outcomes, and, not surprisingly, higher BMI increased the risk to develop atelectasis obscuring target.

Despite dealing with small lesions near the pleura, with only 21% having bronchus sign, 86% in lower lobes, and exclusively located in dependent areas, the procedural outcomes of robotic bronchoscopy with m-CBCT guidance in the LADS group (TIL and DY) were similar to those previously reported with the same technologies in less challenging populations (mostly upper lobes nodules which are less subject to movement and atelectasis, and associated with increased DY).^{8,18-22} In

TABLE 2] Procedural Characteristics

Procedural Characteristics	VESPA Group (n = 33)	LADS Group (n = 29)	P Value
Tidal volume, mL/kg of IBW	7 (6-7.6)	6 (6-7)	.038
PEEP, cm H ₂ O	8 (8-10)	5 (4-5)	< .001
F _{IO₂} , %	60 (46-70)	100 (100-100)	< .001
Intubation to docking time, min	10 (8-15)	18 (15-28)	< .001
Docking to undocking time, min	45 (30-51)	40 (28-62)	.949
Intubation to extubation time, min	76 (65-96)	88 (78-124)	.125
Integration software use	26 (78.8)	26 (89.7)	.312
EBUS staging	17 (51.5)	18 (62.1)	.403
Fluoroscopy time, min	7.9 (6.5-10.1)	8.5 (6.8-9.8)	.631
No. of m-CBCT spins	3 (2-5)	3 (2-6)	.691
Dose area product, Gy·cm ²	56.8 (48.8-100.7)	57.6 (41.9-114.2)	.834
Fluoroscopy dose area product, Gy·cm ²	28.3 (24.2-36.7)	28.8 (17.9-48)	.855
m-CBCT dose area product, Gy·cm ²	28.1 (19.4-61.4)	29.2 (18.7-55.4)	.745
TBNA			
No. of patients	33 (100)	29 (100)	> .999
No. of TBNAs	5.5 (5.0-7.0)	6 (5-6)	.981
Cryobiopsy			
No. of patients	16 (48.5)	12 (41.4)	.575
No. of biopsies	4 (3-4)	3 (3-3.5)	.062
Forceps biopsy			
No. of patients	2 (6.1)	2 (6.9)	1.000
No. of biopsies	5 (4-6)	6 (5-7)	NA
Cytology brush			
No. of patients	3 (9.1)	2 (6.9)	1.000
No. of brushes	2 (1-4)	1.5 (1-2)	NA

Data are presented as No. (%), median (interquartile range), or as otherwise indicated. IBW = ideal body weight; EBUS = endobronchial ultrasound; LADS = lateral decubitus strategy; m-CBCT = mobile cone-beam CT; NA = not available; PEEP = positive end-expiratory pressure; TBNA = trans-bronchial needle aspiration; VESPA = ventilatory strategy to prevent atelectasis.

addition, we found no differences between the 2 groups in terms of time required to perform robotic bronchoscopy, fluoroscopy time, number of m-CBCT spins, radiation, and safety, consistent with previously reported data.^{8,16,18-22} Based on the aforementioned information, in our opinion, having patients in lateral decubitus position does not impair in any fashion the ability to successfully perform shape-sensing robotic bronchoscopy with m-CBCT guidance, and, in fact, it improves its outcomes if nodules are in dependent areas.

As expected, time from intubation to docking of the robot was longer in the LADS group (18 minutes; IQR, 15-28 vs 10 minutes; IQR, 8-15 in the VESPA group; $P < .001$). This prolonged time is not only due to the maneuvers required to position and secure the patient in lateral decubitus, but also due to more challenging

isocentering of the m-CBCT C-arm and collision check, which can be particularly difficult in larger patients with lesions located laterally. That said, our previously reported time from intubation to the first m-CBCT spin when we first described LADS was 52 minutes (range, 45 to 61 minutes).¹² Because registration and navigation only take us a few minutes, we can safely assume that with practice we have cut down the time from intubation to docking of the robot by roughly one-half (to a median of 18 minutes).

The results of the current trial cannot be compared with other ventilatory strategies previously described, such as lung navigation ventilation protocol or lung navigation ventilation protocol plus apneic breath hold, for multiple reasons.^{10,11} The most relevant reason is that in the current trial we only included patients with nodules located in areas at very high risk of atelectasis

(dependent areas, > 80% in lower lobes), whereas for example, in lung navigation ventilation protocol with breath hold, only 21% of the patients had lesions in lower lobes. Other reasons that preclude comparison with these 2 ventilatory strategies is that the authors did not use the more conservative ATS/CHEST definition of DY—because it was not available—and their reports are of retrospective nature, with its inherent bias.

Prone positioning (PP) has been described as another positional strategy to avoid atelectasis during peripheral bronchoscopy in a case report.²³ Although this positioning is highly likely to be effective in preventing atelectasis from obscuring targets located in dorsal areas (it is commonly used by interventional radiologists for CT scan-guided percutaneous approach), when

compared with LADS, it appears to be more cumbersome and potentially riskier. In addition, although LADS resulted in no atelectasis whatsoever in the target lung, in PP, atelectasis is shifted ventrally, and it can still cause CT-to-body divergence. PP requires experienced staff, which may be present in the ICU, but not in the bronchoscopy laboratory. If complications such as bleeding, pneumothorax, or accidental extubation were to arise, these would all be harder to manage.

Our study was carried out in a single institution, which can be seen as a limitation. However, procedures were carried out by 7 different operators in 3 different hospitals (belonging to the same institution). Due to this study's small sample size, a larger trial may be

TABLE 3] Diagnosis on Index Biopsies

VESPA Group (n = 33)	No.	LADS Group (n = 29)	No.
Malignant (n = 16)		Malignant (n = 17)	
Lung adenocarcinoma	7	Lung squamous cell carcinoma	5
Prostate	3	Lung adenocarcinoma	4
Carcinoid	2	Lung large neuroendocrine	1
Ovarian carcinoma	1	Carcinoid	1
Renal cell carcinoma	1	Small cell lung cancer	1
Papillary thyroid carcinoma	1	Breast cancer	2
Endometrial carcinoma	1	Endometrial carcinoma	1
		Prostate	1
		Melanoma	1
Benign (n = 3)		Benign (n = 8)	
Aspergillomas	1	Aspergillomas	2
Mycobacterium avium complex	1	Mycobacterium avium complex	2
Necrotizing granulomatous inflammation	1	Necrotizing granulomatous inflammation	3
		Coccidioides	1
Nondiagnostic samples (n = 5)		Nondiagnostic Samples (n = 4)	
Blood and bronchial cells	2	Blood and bronchial cells	3
Acute inflammation	2	Acute inflammation	1
Necrosis	1		
Diagnosis post-cross over (n = 9)			
Lung adenocarcinoma	2		
Lung squamous cell carcinoma	1		
Non-small cell lung cancer	1		
Histoplasmosis	1		
Hemophilus influenza pneumonia	1		
Necrotizing granulomatous inflammation	1		
Acute inflammation	1		
Blood and bronchial cells	1		

needed to reaffirm the superiority of LADS. Major limitations include the lack of blinding (for obvious reasons), but more importantly, the fact that the assessment of atelectasis obscuring target was left to the bronchoscopist's interpretation of the m-CBCT scan with potential subjectivity. This primary outcome should have ideally been adjudicated in a blinded fashion by independent reviewers to reassure readers about reproducibility and validity. Unfortunately, a post hoc adjudication analysis could not be carried out due to some images being erased from our server. Our study only included patients with lesions in dependent areas, which represent a minority of the patients that we typically see as evident in [Figure 1](#). However, these are the ones at risk of atelectasis obscuring target, and hence our decision when designing the trial. Pure ground-glass opacity lesions were excluded because they are difficult to visualize with m-CBCT scan even without atelectasis. We also excluded processes that cause compressive atelectasis (pleural effusions, ascites, and diaphragmatic paralysis) and severe air-trapping—which in our experience prevents atelectasis—which if unevenly distributed in both arms could bias the results. Finally, although we have personally used

LADS with other robotic platforms, the results of this study cannot be entirely extrapolated.

Interpretation

Our results show that LADS is superior to VESPA in preventing atelectasis from obscuring targets in patients with nodules located in dependent areas, can help eliminate atelectasis, is safe, and is also associated with better procedural outcomes such as TIL and DY in this population.

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